IT 340: BUSINESS INTELLIGENCE SYSTEMS

3.3 Sample Files – Basic SQL Functions

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September 2026

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1. Adding Employee Manager Information

The Employee table contains a ReportsTo column that identifies the employee's manager. Since the manager is also an employee, I used a self-join. Because the question specifies two layers of management, the query identifies the employee, the employee's immediate manager, and the manager's manager.

SQL Query

```
 SELECT
    e.FirstName AS EmployeeFirstName,
    e.LastName AS EmployeeLastName,
    m.FirstName AS ManagerFirstName,
    m.LastName AS ManagerLastName,
    gm.FirstName AS ManagerManagerFirstName,
    gm.LastName AS ManagerManagerLastName
FROM Employee e
LEFT JOIN Employee m
    ON e.ReportsTo = m.EmployeeId
LEFT JOIN Employee gm
    ON m.ReportsTo = gm.EmployeeId
ORDER BY gm.LastName, m.LastName, e.LastName;
```

Explanation

The first LEFT JOIN connects each employee to their immediate manager. The second LEFT JOIN connects that manager to the next level of management. This provides two layers of management while keeping employees who may not have a manager in the results.

2. Which Artists Have Produced the Most Songs?

To answer this question, I used the relationship between the Artist, Album, and Track tables. An artist can have multiple albums, and each album can contain multiple tracks. Therefore,

counting tracks associated with each artist provides a direct measure of the number of songs represented in the database.

SQL Query

```
SELECT
    ar.Name AS Artist,
    COUNT(t.TrackId) AS NumberOfSongs
FROM Artist ar
JOIN Album al
    ON ar.ArtistId = al.ArtistId
JOIN Track t
    ON al.AlbumId = t.AlbumId
GROUP BY ar.ArtistId, ar.Name
ORDER BY NumberOfSongs DESC
LIMIT 10;
```

Explanation

The query groups the tracks by artist and orders the results from the artist with the greatest number of songs to the lowest.

Running this query against the Chinook database shows that Iron Maiden has produced the most songs, with 213 tracks which is well ahead of the next closest artist, U2, with 135.

Defensible Position

Metric Proposed

I propose using the number of tracks associated with each artist as the metric for determining which artists have produced the most songs.

Three Reasons This Is a Good Metric

1. **It directly answers the manager's question.**
   The manager wants to know which artists have produced the most songs, so counting the number of tracks directly measures the quantity of songs associated with each artist.

2. **Catalog depth:** It acts as a reliable proxy for catalog depth. In a digital storefront, knowing which artists have the largest volume of tracks helps identify where the bulk of the store's inventory originates.

3. **It allows artists to be ranked objectively.**
   Each artist receives a numerical track count, making it possible to compare artists and identify those with the largest number of songs.

Three Limitations of the Metric

1. **It measures quantity rather than popularity.**
   An artist with more songs in the database is not necessarily more popular or successful than an artist with fewer songs.

2. **It does not measure sales or revenue.**
   The number of tracks does not show how many times the songs were purchased or how much revenue they generated.

3. **It does not measure song quality or customer engagement.**
   The metric treats every track equally and does not consider customer ratings, listening behavior, purchases, or other measures of audience engagement.

Additional Data I Would Like to Use

To improve the metric, I would like to have sales and customer-engagement data for each artist and track. This could include the number of times each track was purchased, total revenue generated, quantity sold, customer ratings, and streaming or play counts.

This additional information would allow the analysis to move beyond simply identifying which artists have the most songs and provide a better understanding of which artists are actually the most successful with customers.